

CREDIT CARD TRANSACTION RISK FLAGGING SYSTEM

Analyst Risk Report | Full Year 2023 | MySQL Risk Analytics Database

Executive Risk Summary

Total Transactions	High-Value Anomalies	Risk-Flagged Transactions	High-Risk Customers	CRITICAL Alerts
12,000	15+ (showing top 15)	600	12	15

This report flags transactions and entities that triggered one or more of the 5 risk detection rules: High-Value Anomaly (3× baseline spend), Velocity Check (5+ txns/24h), Repeated Failure Pattern (3+ failures/month), Credit Limit Breach (≥80% of limit), and Merchant Anomaly. All alerts require human analyst review before any action is taken.

Rule 1 — High-Value Anomaly Alerts (Top 15 by Spend Multiplier)

Txn ID	Customer	Merchant	Date	Txn Amt (■)	Avg Spend (■)	Multiplier	Level	Category
TXN0004693	C00263	M0051	2023-05-22	■141,417	■2,416	58.53×	CRITICAL	Retail
TXN0010059	C00445	M0055	2023-11-01	■166,967	■4,422	37.76×	CRITICAL	Healthcare
TXN0001822	C00436	M0106	2023-02-24	■188,134	■5,227	35.99×	CRITICAL	Food & Beverage
TXN0004330	C00679	M0025	2023-05-10	■157,297	■4,387	35.86×	CRITICAL	Electronics
TXN0010788	C00270	M0113	2023-11-24	■146,794	■4,473	32.81×	CRITICAL	Healthcare
TXN0008402	C00307	M0048	2023-09-12	■159,299	■5,401	29.49×	CRITICAL	Food & Beverage
TXN0004276	C00651	M0005	2023-05-09	■170,589	■5,815	29.33×	CRITICAL	Healthcare
TXN0011256	C00751	M0060	2023-12-08	■178,764	■6,242	28.64×	CRITICAL	Food & Beverage
TXN0007102	C00046	M0008	2023-08-04	■149,246	■6,314	23.64×	CRITICAL	Electronics
TXN0002730	C00185	M0044	2023-03-23	■172,978	■7,395	23.39×	CRITICAL	Healthcare
TXN0001322	C00132	M0058	2023-02-09	■194,180	■8,321	23.34×	CRITICAL	Entertainment
TXN0006331	C00039	M0089	2023-07-12	■181,448	■8,058	22.52×	CRITICAL	Travel
TXN0002002	C00528	M0106	2023-03-02	■129,504	■5,824	22.24×	CRITICAL	Food & Beverage
TXN0010520	C00241	M0033	2023-11-15	■190,924	■9,232	20.68×	CRITICAL	Food & Beverage
TXN0004060	C00350	M0064	2023-05-02	■99,844	■4,946	20.18×	CRITICAL	Fuel

Rule 3 — Repeated Failure Pattern (Top 10)

Customer ID	Name	Segment	Month	Failed Txns	Total Attempted (■)
C00058	Ritika Kumar	Corporate	2023-10	3	■50,354
C00217	Rohit Chauhan	Premium	2023-06	3	■7,538
C00418	Sneha Shah	Retail	2023-03	3	■105,449
C00534	Gaurav Patel	Corporate	2023-02	3	■4,602
C00747	Gaurav Kumar	Premium	2023-12	3	■7,300

Rule 4 — Credit Limit Breach Proximity (Top 10)

Txn ID	Customer ID	Name	Credit Limit	Txn Amount	% of Limit	Status	Date
TXN0005304	C00782	Nisha Kapoor	■25,000	■199,270	797.1%	Success	2023-06-10
TXN0009042	C00195	Sneha Malhotra	■25,000	■194,461	777.8%	Failed	2023-10-02
TXN0002289	C00677	Divya Chauhan	■25,000	■193,433	773.7%	Success	2023-03-10
TXN0007971	C00734	Preeti Gupta	■25,000	■193,205	772.8%	Success	2023-08-30
TXN0004842	C00759	Swati Pandey	■25,000	■190,593	762.4%	Success	2023-05-26
TXN0004040	C00124	Meera Kumar	■25,000	■188,651	754.6%	Disputed	2023-05-01
TXN0006489	C00371	Suresh Rao	■25,000	■188,453	753.8%	Failed	2023-07-17
TXN0006757	C00508	Sneha Nair	■25,000	■188,450	753.8%	Success	2023-07-25
TXN0006465	C00195	Sneha Malhotra	■25,000	■188,261	753.0%	Success	2023-07-16
TXN0002188	C00032	Ritika Sharma	■25,000	■188,086	752.3%	Success	2023-03-07

High-Risk Customer Watchlist

Customer ID	Name	Segment	Total Txns	Failed	Risk Flagged	Avg Spend	Fail Rate	Risk
C00482	Priya Mishra	Premium	30	6	3	■23,131	20.0%	HIGH RISK
C00139	Deepak Iyer	Retail	22	4	3	■21,779	18.2%	HIGH RISK
C00195	Sneha Malhotra	Retail	17	4	4	■25,961	23.5%	HIGH RISK
C00217	Rohit Chauhan	Premium	18	4	3	■35,419	22.2%	HIGH RISK
C00412	Sneha Reddy	Corporate	20	4	4	■34,097	20.0%	HIGH RISK
C00019	Pooja Mishra	Corporate	27	3	3	■28,710	11.1%	HIGH RISK
C00223	Arun Malhotra	Corporate	19	3	4	■26,323	15.8%	HIGH RISK
C00433	Amit Shah	Corporate	23	3	3	■36,312	13.0%	HIGH RISK
C00519	Sanjay Patel	Retail	17	3	3	■21,260	17.6%	HIGH RISK
C00656	Pooja Shah	Retail	18	3	3	■26,907	16.7%	HIGH RISK
C00697	Rahul Shah	Retail	18	3	3	■29,646	16.7%	HIGH RISK
C00191	Swati Tiwari	Corporate	13	2	4	■55,658	15.4%	HIGH RISK

Analyst Recommendations

1. **CRITICAL High-Value Alerts:** All CRITICAL-level transactions with spend multiplier >10× should be reviewed within 24 hours. Customer accounts should be temporarily flagged pending verification.

2. **High-Risk Customer Watchlist:** These customers show persistent failure patterns or multiple risk-flagged transactions. Recommend enhanced authentication and a 30-day monitoring window.

3. **Credit Limit Breaches:** Several customers attempted transactions exceeding 600–800% of their credit limit — these require immediate review by the risk team.

4. **Velocity Alerts:** The velocity detection rule (5+ txns/24h) is a leading indicator for account takeover. Consider real-time alerting at the payment gateway layer.

5. **Run Nightly:** Execute CALL sp_populate_risk_alerts() on a nightly schedule to keep the risk_alerts table current for the compliance dashboard.

